

Consider the training of the following multilayer neural network on the operation of XOR.

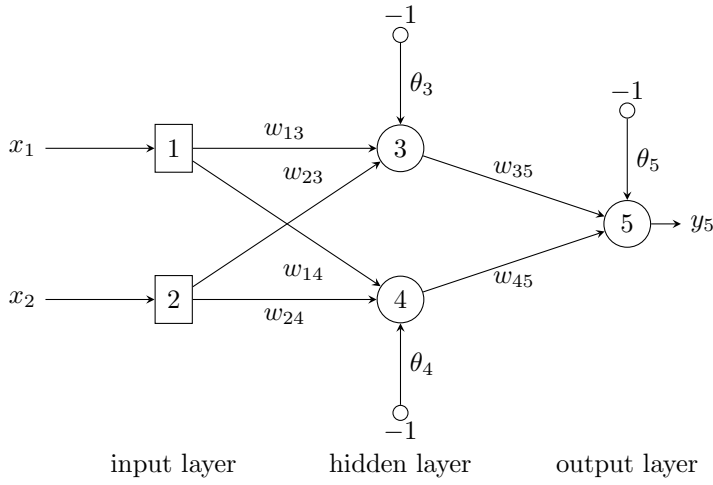


Table 1: XOR operation

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

1 Initialisation

α	w_{13}	w_{23}	w_{14}	w_{24}	w_{35}	w_{45}	θ_3	θ_4	θ_5
0.5	0.1	0.2	0.15	0.2	0.1	0.1	0.2	0.3	0.15

2 Epoch 1

2.1 Iteration 1

- Inputs: $x_1 = 0$ and $x_2 = 0$
- Expected/desired output: $y_d = 0$

$$\begin{aligned} \text{output of hidden neuron 3, } y_3 &= \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\ &= \text{sigmoid}(0.1 \cdot 0 + 0.2 \cdot 0 - 0.2) = 0.4501 \end{aligned}$$

$$\begin{aligned} \text{output of hidden neuron 4, } y_4 &= \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\ &= \text{sigmoid}(0.15 \cdot 0 + 0.2 \cdot 0 - 0.3) = 0.4256 \end{aligned}$$

$$\begin{aligned} \text{output of output neuron 5, } y_5 &= \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\ &= \text{sigmoid}(0.1 \cdot 0.45013 + 0.1 \cdot 0.42555 - 0.15) = 0.4844 \end{aligned}$$

$$\text{error} = -0.4844$$

$$\begin{aligned} \text{error gradient for neuron 5, } \delta_5 &= y_5(1 - y_5)e_5 \\ &= 0.4844 \cdot (1 - 0.4844) \cdot -0.4844 = -0.121 \end{aligned}$$

$$\begin{aligned} \text{error gradient for neuron 3, } \delta_3 &= y_3(1 - y_3)(\delta_5 w_{35}) \\ &= 0.4501 \cdot (1 - 0.4501) \cdot (-0.121 \cdot 0.1) = -0.003 \end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4501 \cdot -0.121 = -0.0272 \\
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4256 \cdot -0.121 = -0.0257 \\
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot -0.003 = 0 \\
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot -0.003 = 0 \\
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot -0.0029 = 0 \\
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot -0.0029 = 0
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.1 + 0 = 0.1 \\
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.2 + 0 = 0.2 \\
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.15 + 0 = 0.15 \\
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.2 + 0 = 0.2 \\
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.1 + -0.0272 = 0.0728 \\
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.1 + -0.0257 = 0.0743
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta \theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot -0.003 = 0.0015 \\
&\text{threshold correction for neuron 4, } \Delta \theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot -0.0029 = 0.0015 \\
&\text{threshold correction for neuron 5, } \Delta \theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot -0.121 = 0.0605
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta \theta_3 \\
&\quad = 0.2 + 0.0015 = 0.2015 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta \theta_4 \\
&\quad = 0.3 + 0.0015 = 0.3015 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta \theta_5 \\
&\quad = 0.15 + 0.0605 = 0.2105
\end{aligned}$$

2.2 Iteration 2

- Inputs: $x_1 = 0$ and $x_2 = 1$
- Expected/desired output: $y_d = 1$

$$\begin{aligned}
&\text{output of hidden neuron 3, } y_3 = \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\
&\quad = \text{sigmoid}(0.1 \cdot 0 + 0.2 \cdot 1 - 0.2015) = 0.4996 \\
&\text{output of hidden neuron 4, } y_4 = \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\
&\quad = \text{sigmoid}(0.15 \cdot 0 + 0.2 \cdot 1 - 0.3015) = 0.4746 \\
&\text{output of output neuron 5, } y_5 = \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\
&\quad = \text{sigmoid}(0.0728 \cdot 0.4996 + 0.0743 \cdot 0.4746 - 0.2105) = 0.4653
\end{aligned}$$

$$\text{error} = 0.5347$$

$$\begin{aligned}
&\text{error gradient for neuron 5, } \delta_5 = y_5(1 - y_5)e_5 \\
&\quad = 0.4653 \cdot (1 - 0.4653) \cdot 0.5347 = 0.133
\end{aligned}$$

$$\begin{aligned}
&\text{error gradient for neuron 3, } \delta_3 = y_3(1 - y_3)(\delta_5 w_{35}) \\
&\quad = 0.4996 \cdot (1 - 0.4996) \cdot (0.133 \cdot 0.07278) = 0.0024
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4996 \cdot 0.133 = 0.0332
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4746 \cdot 0.133 = 0.0316
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot 0.0024 = 0
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot 0.0024 = 0.0012
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot 0.0025 = 0
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot 0.0025 = 0.0012
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.1 + 0 = 0.1
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.2 + 0.0012 = 0.2012
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.15 + 0 = 0.15
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.2 + 0.0012 = 0.2012
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.07278 + 0.0332 = 0.106
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.07428 + 0.0316 = 0.1058
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta \theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0024 = -0.0012
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 4, } \Delta \theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0025 = -0.0012
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 5, } \Delta \theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot 0.133 = -0.0665
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta\theta_3 \\
&\quad = 0.20149 + -0.0012 = 0.2003 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta\theta_4 \\
&\quad = 0.30147 + -0.0012 = 0.3003 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta\theta_5 \\
&\quad = 0.21048 + -0.0665 = 0.144
\end{aligned}$$

2.3 Iteration 3

- Inputs: $x_1 = 1$ and $x_2 = 0$
- Expected/desired output: $y_d = 1$

$$\begin{aligned}
&\text{output of hidden neuron 3, } y_3 = \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\
&\quad = \text{sigmoid}(0.1 \cdot 1 + 0.2012 \cdot 0 - 0.2003) = 0.4749 \\
&\text{output of hidden neuron 4, } y_4 = \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\
&\quad = \text{sigmoid}(0.15 \cdot 1 + 0.2012 \cdot 0 - 0.3003) = 0.4625 \\
&\text{output of output neuron 5, } y_5 = \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\
&\quad = \text{sigmoid}(0.106 \cdot 0.4749 + 0.1058 \cdot 0.4625 - 0.144) = 0.4888
\end{aligned}$$

$$\text{error} = 0.5112$$

$$\begin{aligned}
&\text{error gradient for neuron 5, } \delta_5 = y_5(1 - y_5)e_5 \\
&\quad = 0.4888 \cdot (1 - 0.4888) \cdot 0.5112 = 0.1277 \\
&\text{error gradient for neuron 3, } \delta_3 = y_3(1 - y_3)(\delta_5 w_{35}) \\
&\quad = 0.4749 \cdot (1 - 0.4749) \cdot (0.1277 \cdot 0.106) = 0.0034
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4749 \cdot 0.1277 = 0.0303 \\
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4625 \cdot 0.1277 = 0.0295 \\
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot 0.0034 = 0.0017 \\
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot 0.0034 = 0 \\
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot 0.0033 = 0.0017 \\
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot 0.0033 = 0
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.1 + 0.0017 = 0.1017 \\
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.2012 + 0 = 0.2012 \\
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.15 + 0.0017 = 0.1517 \\
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.20122 + 0 = 0.2012 \\
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.106 + 0.0303 = 0.1363 \\
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.10583 + 0.0295 = 0.1354
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta\theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0034 = -0.0017 \\
&\text{threshold correction for neuron 4, } \Delta\theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0033 = -0.0017 \\
&\text{threshold correction for neuron 5, } \Delta\theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot 0.1277 = -0.0639
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta\theta_3 \\
&\quad = 0.20029 + -0.0017 = 0.1986 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta\theta_4 \\
&\quad = 0.30025 + -0.0017 = 0.2986 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta\theta_5 \\
&\quad = 0.14397 + -0.0639 = 0.0801
\end{aligned}$$

2.4 Iteration 4

- Inputs: $x_1 = 1$ and $x_2 = 1$
- Expected/desired output: $y_d = 0$

$$\begin{aligned}
&\text{output of hidden neuron 3, } y_3 = \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\
&\quad = \text{sigmoid}(0.1017 \cdot 1 + 0.2012 \cdot 1 - 0.1986) = 0.526 \\
&\text{output of hidden neuron 4, } y_4 = \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\
&\quad = \text{sigmoid}(0.1517 \cdot 1 + 0.2012 \cdot 1 - 0.2986) = 0.5135 \\
&\text{output of output neuron 5, } y_5 = \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\
&\quad = \text{sigmoid}(0.1363 \cdot 0.526 + 0.1354 \cdot 0.5135 - 0.0801) = 0.5152
\end{aligned}$$

$$\text{error} = -0.5152$$

$$\begin{aligned}
&\text{error gradient for neuron 5, } \delta_5 = y_5(1 - y_5)e_5 \\
&\quad = 0.5152 \cdot (1 - 0.5152) \cdot -0.5152 = -0.1287 \\
&\text{error gradient for neuron 3, } \delta_3 = y_3(1 - y_3)(\delta_5 w_{35}) \\
&\quad = 0.526 \cdot (1 - 0.526) \cdot (-0.1287 \cdot 0.13632) = -0.0044
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.526 \cdot -0.1287 = -0.0338 \\
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.5135 \cdot -0.1287 = -0.033 \\
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022 \\
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022 \\
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022 \\
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.10168 + -0.0022 = 0.0995 \\
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.2012 + -0.0022 = 0.199 \\
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.15166 + -0.0022 = 0.1495 \\
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.20122 + -0.0022 = 0.1991 \\
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.13632 + -0.0338 = 0.1025 \\
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.13536 + -0.033 = 0.1023
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta \theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot -0.0044 = 0.0022 \\
&\text{threshold correction for neuron 4, } \Delta \theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot -0.0044 = 0.0022 \\
&\text{threshold correction for neuron 5, } \Delta \theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot -0.1287 = 0.0643
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta \theta_3 \\
&\quad = 0.19861 + 0.0022 = 0.2008 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta \theta_4 \\
&\quad = 0.29858 + 0.0022 = 0.3008 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta \theta_5 \\
&\quad = 0.08011 + 0.0643 = 0.1444
\end{aligned}$$

3 Epoch 2

3.1 Iteration 1

- Inputs: $x_1 = 0$ and $x_2 = 0$
- Expected/desired output: $y_d = 0$

$$\begin{aligned}
&\text{output of hidden neuron 3, } y_3 = \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\
&\quad = \text{sigmoid}(0.0995 \cdot 0 + 0.199 \cdot 0 - 0.2008) = 0.45 \\
&\text{output of hidden neuron 4, } y_4 = \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\
&\quad = \text{sigmoid}(0.1495 \cdot 0 + 0.1991 \cdot 0 - 0.3008) = 0.4254 \\
&\text{output of output neuron 5, } y_5 = \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\
&\quad = \text{sigmoid}(0.1025 \cdot 0.45 + 0.1023 \cdot 0.4254 - 0.1444) = 0.4862
\end{aligned}$$

$$\text{error} = -0.4862$$

$$\begin{aligned}
&\text{error gradient for neuron 5, } \delta_5 = y_5(1 - y_5)e_5 \\
&\quad = 0.4862 \cdot (1 - 0.4862) \cdot -0.4862 = -0.1214 \\
&\text{error gradient for neuron 3, } \delta_3 = y_3(1 - y_3)(\delta_5 w_{35}) \\
&\quad = 0.45 \cdot (1 - 0.45) \cdot (-0.1214 \cdot 0.1025) = -0.0031
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.45 \cdot -0.1214 = -0.0273 \\
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4254 \cdot -0.1214 = -0.0258 \\
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot -0.0031 = 0 \\
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot -0.0031 = 0 \\
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot -0.003 = 0 \\
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot -0.003 = 0
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.0995 + 0 = 0.0995 \\
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.19902 + 0 = 0.199 \\
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.14949 + 0 = 0.1495 \\
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.19905 + 0 = 0.1991 \\
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.1025 + -0.0273 = 0.0752 \\
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.10234 + -0.0258 = 0.0765
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta \theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot -0.0031 = 0.0015 \\
&\text{threshold correction for neuron 4, } \Delta \theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot -0.003 = 0.0015 \\
&\text{threshold correction for neuron 5, } \Delta \theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot -0.1214 = 0.0607
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta\theta_3 \\
&\quad = 0.20079 + 0.0015 = 0.2023 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta\theta_4 \\
&\quad = 0.30075 + 0.0015 = 0.3023 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta\theta_5 \\
&\quad = 0.14444 + 0.0607 = 0.2052
\end{aligned}$$

3.2 Iteration 2

- Inputs: $x_1 = 0$ and $x_2 = 1$
- Expected/desired output: $y_d = 1$

$$\begin{aligned}
&\text{output of hidden neuron 3, } y_3 = \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\
&\quad = \text{sigmoid}(0.0995 \cdot 0 + 0.199 \cdot 1 - 0.2023) = 0.4991 \\
&\text{output of hidden neuron 4, } y_4 = \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\
&\quad = \text{sigmoid}(0.1495 \cdot 0 + 0.1991 \cdot 1 - 0.3023) = 0.4742 \\
&\text{output of output neuron 5, } y_5 = \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\
&\quad = \text{sigmoid}(0.0752 \cdot 0.4991 + 0.0765 \cdot 0.4742 - 0.2052) = 0.4672
\end{aligned}$$

$$\text{error} = 0.5328$$

$$\begin{aligned}
&\text{error gradient for neuron 5, } \delta_5 = y_5(1 - y_5)e_5 \\
&\quad = 0.4672 \cdot (1 - 0.4672) \cdot 0.5328 = 0.1326 \\
&\text{error gradient for neuron 3, } \delta_3 = y_3(1 - y_3)(\delta_5 w_{35}) \\
&\quad = 0.4991 \cdot (1 - 0.4991) \cdot (0.1326 \cdot 0.07518) = 0.0025
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4991 \cdot 0.1326 = 0.0331 \\
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4742 \cdot 0.1326 = 0.0314 \\
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot 0.0025 = 0 \\
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot 0.0025 = 0.0012 \\
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot 0.0025 = 0 \\
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot 0.0025 = 0.0013
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.0995 + 0 = 0.0995 \\
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.19902 + 0.0012 = 0.2003 \\
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.14949 + 0 = 0.1495 \\
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.19905 + 0.0013 = 0.2003 \\
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.07518 + 0.0331 = 0.1083 \\
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.07652 + 0.0314 = 0.108
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta\theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0025 = -0.0012 \\
&\text{threshold correction for neuron 4, } \Delta\theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0025 = -0.0013 \\
&\text{threshold correction for neuron 5, } \Delta\theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot 0.1326 = -0.0663
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta\theta_3 \\
&\quad = 0.20232 + -0.0012 = 0.2011 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta\theta_4 \\
&\quad = 0.30226 + -0.0013 = 0.301 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta\theta_5 \\
&\quad = 0.20515 + -0.0663 = 0.1389
\end{aligned}$$

3.3 Iteration 3

- Inputs: $x_1 = 1$ and $x_2 = 0$
- Expected/desired output: $y_d = 1$

$$\begin{aligned}
&\text{output of hidden neuron 3, } y_3 = \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\
&\quad = \text{sigmoid}(0.0995 \cdot 1 + 0.2003 \cdot 0 - 0.2011) = 0.4746 \\
&\text{output of hidden neuron 4, } y_4 = \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\
&\quad = \text{sigmoid}(0.1495 \cdot 1 + 0.2003 \cdot 0 - 0.301) = 0.4622 \\
&\text{output of output neuron 5, } y_5 = \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\
&\quad = \text{sigmoid}(0.1083 \cdot 0.4746 + 0.108 \cdot 0.4622 - 0.1389) = 0.4906
\end{aligned}$$

$$\text{error} = 0.5094$$

$$\begin{aligned}
&\text{error gradient for neuron 5, } \delta_5 = y_5(1 - y_5)e_5 \\
&\quad = 0.4906 \cdot (1 - 0.4906) \cdot 0.5094 = 0.1273 \\
&\text{error gradient for neuron 3, } \delta_3 = y_3(1 - y_3)(\delta_5 w_{35}) \\
&\quad = 0.4746 \cdot (1 - 0.4746) \cdot (0.1273 \cdot 0.10826) = 0.0034
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4746 \cdot 0.1273 = 0.0302 \\
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.4622 \cdot 0.1273 = 0.0294 \\
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot 0.0034 = 0.0017 \\
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 0 \cdot 0.0034 = 0 \\
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot 0.0034 = 0.0017 \\
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 0 \cdot 0.0034 = 0
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.0995 + 0.0017 = 0.1012 \\
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.20026 + 0 = 0.2003 \\
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.14949 + 0.0017 = 0.1512 \\
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.2003 + 0 = 0.2003 \\
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.10826 + 0.0302 = 0.1385 \\
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.10796 + 0.0294 = 0.1374
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta \theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0034 = -0.0017 \\
&\text{threshold correction for neuron 4, } \Delta \theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot 0.0034 = -0.0017 \\
&\text{threshold correction for neuron 5, } \Delta \theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot 0.1273 = -0.0636
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta \theta_3 \\
&\quad = 0.20108 + -0.0017 = 0.1994 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta \theta_4 \\
&\quad = 0.30101 + -0.0017 = 0.2993 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta \theta_5 \\
&\quad = 0.13885 + -0.0636 = 0.0752
\end{aligned}$$

3.4 Iteration 4

- Inputs: $x_1 = 1$ and $x_2 = 1$
- Expected/desired output: $y_d = 0$

$$\begin{aligned}
&\text{output of hidden neuron 3, } y_3 = \text{sigmoid}(w_{13}x_1 + w_{23}x_2 - \theta_3) \\
&\quad = \text{sigmoid}(0.1012 \cdot 1 + 0.2003 \cdot 1 - 0.1994) = 0.5254 \\
&\text{output of hidden neuron 4, } y_4 = \text{sigmoid}(w_{14}x_1 + w_{24}x_2 - \theta_4) \\
&\quad = \text{sigmoid}(0.1512 \cdot 1 + 0.2003 \cdot 1 - 0.2993) = 0.513 \\
&\text{output of output neuron 5, } y_5 = \text{sigmoid}(w_{35}y_3 + w_{45}y_4 - \theta_5) \\
&\quad = \text{sigmoid}(0.1385 \cdot 0.5254 + 0.1374 \cdot 0.513 - 0.0752) = 0.5169
\end{aligned}$$

$$\text{error} = -0.5169$$

$$\begin{aligned}
&\text{error gradient for neuron 5, } \delta_5 = y_5(1 - y_5)e_5 \\
&\quad = 0.5169 \cdot (1 - 0.5169) \cdot -0.5169 = -0.1291 \\
&\text{error gradient for neuron 3, } \delta_3 = y_3(1 - y_3)(\delta_5 w_{35}) \\
&\quad = 0.5254 \cdot (1 - 0.5254) \cdot (-0.1291 \cdot 0.13846) = -0.0044
\end{aligned}$$

$$\begin{aligned}
&\text{weight correction for } w_{35}, \Delta w_{35} = \alpha \times y_3 \times \delta_5 \\
&\quad = 0.5 \cdot 0.5254 \cdot -0.1291 = -0.0339 \\
&\text{weight correction for } w_{45}, \Delta w_{45} = \alpha \times y_4 \times \delta_5 \\
&\quad = 0.5 \cdot 0.513 \cdot -0.1291 = -0.0331 \\
&\text{weight correction for } w_{13}, \Delta w_{13} = \alpha \times y_1 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022 \\
&\text{weight correction for } w_{23}, \Delta w_{23} = \alpha \times y_2 \times \delta_3 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022 \\
&\text{weight correction for } w_{14}, \Delta w_{14} = \alpha \times y_1 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022 \\
&\text{weight correction for } w_{24}, \Delta w_{24} = \alpha \times y_2 \times \delta_4 \\
&\quad = 0.5 \cdot 1 \cdot -0.0044 = -0.0022
\end{aligned}$$

$$\begin{aligned}
&\text{new weight for } w_{13}, w_{13} = w_{13} + \Delta w_{13} \\
&\quad = 0.10121 + -0.0022 = 0.099 \\
&\text{new weight for } w_{23}, w_{23} = w_{23} + \Delta w_{23} \\
&\quad = 0.20026 + -0.0022 = 0.198 \\
&\text{new weight for } w_{14}, w_{14} = w_{14} + \Delta w_{14} \\
&\quad = 0.15118 + -0.0022 = 0.149 \\
&\text{new weight for } w_{24}, w_{24} = w_{24} + \Delta w_{24} \\
&\quad = 0.2003 + -0.0022 = 0.1981 \\
&\text{new weight for } w_{35}, w_{35} = w_{35} + \Delta w_{35} \\
&\quad = 0.13846 + -0.0339 = 0.1046 \\
&\text{new weight for } w_{45}, w_{45} = w_{45} + \Delta w_{45} \\
&\quad = 0.13736 + -0.0331 = 0.1043
\end{aligned}$$

$$\begin{aligned}
&\text{threshold correction for neuron 3, } \Delta \theta_3 = \alpha \times (-1) \times \delta_3 \\
&\quad = 0.5 \cdot (-1) \cdot -0.0044 = 0.0022 \\
&\text{threshold correction for neuron 4, } \Delta \theta_4 = \alpha \times (-1) \times \delta_4 \\
&\quad = 0.5 \cdot (-1) \cdot -0.0044 = 0.0022 \\
&\text{threshold correction for neuron 5, } \Delta \theta_5 = \alpha \times (-1) \times \delta_5 \\
&\quad = 0.5 \cdot (-1) \cdot -0.1291 = 0.0645
\end{aligned}$$

$$\begin{aligned}
&\text{new threshold for neuron 3, } \theta_3 = \theta_3 + \Delta\theta_3 \\
&\qquad\qquad\qquad = 0.19937 + 0.0022 = 0.2016 \\
&\text{new threshold for neuron 4, } \theta_4 = \theta_4 + \Delta\theta_4 \\
&\qquad\qquad\qquad = 0.29932 + 0.0022 = 0.3015 \\
&\text{new threshold for neuron 5, } \theta_5 = \theta_5 + \Delta\theta_5 \\
&\qquad\qquad\qquad = 0.07521 + 0.0645 = 0.1397
\end{aligned}$$

CSC3034 Current Course Update

MLNN calculation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Lab 6 trains and evaluates a scikit-learn MLP on the Wine dataset.
- Current Lab 7 visualizes neural-network hyperplanes on standardized Iris features.
- Current Lab 8 adds Keras CNN/LSTM examples and Isaac Sim vision demonstrations.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.